

Kai-Po Wei

TEL: +886-965-323-336; Email: kaipowei@umich.edu; Github: <https://github.com/kaipowei>

EDUCATION

M.S. in Robotics

Department of Robotics, University of Michigan

09/2026-05/2028

B.S. in Vehicle Engineering

Department of Vehicle Engineering, National Taipei University of Technology

09/2021-01/2026

- Last 60 GPA: 3.9/4.0, Courses: Vehicle Dynamics, Automatic Control, Statics, Mechanics of Materials

RESEARCH EXPERIENCE

Structural & Optimization Lab, National Taipei University of Technology

Topic: *Scooter Frame Fatigue Life Analysis and Optimization*

09/2023 - 07/2024

- Led a **four-member team** that conducted fatigue analysis on a scooter frame under five loading conditions in accordance with *industry standards*
- Developed CAD models in SolidWorks and performed finite element simulations using HyperWorks
- Evaluated stress distribution and fatigue life, optimizing the frame design for durability (fatigue life increased by 339% with only 12.5% weight increase)

PROJECT EXPERIENCE

AI CUP 2025 – Ministry of Education National Intercollegiate Artificial Intelligence Competition (Taiwan)

Advisor: Prof. Tung-Ju Hsieh

Independent Researcher

9/2025 - 1/2026

- Built two independent end-to-end deep learning pipelines for cardiac CT analysis: trained a **MONAI UNet3D** (DiceCELoss, AdamW) for 3D myocardial segmentation (**0.7439 Dice**) and fine-tuned **YOLOv12** for aortic-valve detection (**0.9396 Dice**)
- Iterated on training configuration (batch size, checkpoint warm-starts) and inference-time confidence/IoU thresholds under a 3-submissions/day limit; handled 3D preprocessing (HU clipping, spacing resampling, ROI extraction)

Taipei Tech Racing FSAE team, National Taipei University of Technology

Chassis team member

10/2022 - 07/2023

- **Second Place**, Formula Student Taiwan 2023
- Designed and optimized suspension geometry to enhance vehicle *handling & stability*
- Designed, analyzed, and manufactured drivetrain components; achieved **31% cost reduction** and **16% weight savings**
- Performed finite element analysis (FEA) to ensure structural integrity of drivetrain parts
- Supported CNC machining and welding-based manufacturing

WORK EXPERIENCE

IPE Innotech Performance Exhaust

Project & Product management, R&D Intern

07/2024 - 08/2024

- Contributed to the design, simulation testing, and manufacturing of three magnesium alloy wheels for Porsche GT3RS
- Increased forged wheel *production efficiency* by **16%** through *workflow redesign*

Foxconn CCEV (Center of Commercial Electric Vehicle)

R&D Intern

07/2023-08/2023

- Participated in the design and manufacturing of a full-electric commercial van- *Model N*
- Contributed to cockpit human-factors engineering; **two proposals adopted into final design**
- Collaborated with cross-functional teams (*ADAS/Chassis/Human Factors*)

SKILLS

- **Software:** SolidWorks, HyperWorks, BikeSim, CarSim
- **Programming:** Python, MATLAB, Simulink, C++, PyTorch, OpenCV, ROS, Git, MONAI, Ultralytics YOLO
- **Engineering:** *Fatigue analysis*, FEA, structural optimization, vehicle dynamics
- **Hands-on:** Mechanical maintenance, repair, machining